E-Commerce Sales & Customer Analysis

SQL Data Analysis Project

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Business Overview

- A growing E-commerce company operating in Egypt, offering a diverse range of products including (Electronics, Accessories, Furniture).
- The company serves customers across multiple cities, enabling them to place orders using various payment methods such as (Cash, Visa, Instapay).
- Each order may include applied discounts and can have different statuses, including (Delivered, Cancelled, Pending).
- The business is expanding and generating increasing volumes of transactional data.

Business Problem

- ❖ Despite the growth, the management team lacks clear visibility into key business areas:
  - Sales performance across products and locations.
  - Customer purchasing behavior and preferences.
  - The effectiveness of discounts on revenue.
  - Order status trends (Delivered vs Cancelled vs Pending).
- ❖ This lack of insight makes it difficult to:
  - Identify growth opportunities.
  - Optimize business strategies.
  - Make informed, data-driven decisions.

Objectives

Understand the business data

Gain a clear understanding of the dataset structure, including orders, customers, products, and transactions.

Analyze sales & performance

Identify sales trends, top-performing products, and revenue distribution across cities and categories.

Explore customer behavior

Identify patterns in purchasing habits, payment preferences, and regional activity.

Generate actionable insights

Translate analysis into actionable insights that support business growth and optimization.

Dataset Description

The dataset contains

- E-commerce transactional dataset.
- Includes Orders, Customers, and Products.
- Covers sales, customer details, and product information.

Data Source

- Data provided as CSV files
- Files include:
  - orders.csv
  - customers.csv
  - products.csv

Data Preparation

- Created database using **SQL Server**
- Imported **CSV files** into tables
- Defined relationships using **primary & foreign keys**
- Prepared data for analysis

Data transformed from flat files into a relational model to enable efficient SQL analysis.

Entity Relationship Diagram (ERD) of the database

Key Insights & Results

Q1 - Retrieve all orders that occurred in 2023.

Results | Messages

	order_id	customer_id	product_id	order_date	quantity	payment_method	order_status	discount
1	1	81	10	2023-10-25	3	Instapay	Delivered	0
2	2	90	7	2023-02-05	1	Cash	Delivered	5
3	3	8	1	2023-03-25	3	Cash	Delivered	0
4	4	93	7	2023-09-20	2	Cash	Delivered	10
5	5	26	10	2023-11-06	4	Cash	Pending	15
6	6	74	5	2023-06-21	1	Cash	Delivered	0
7	7	90	6	2023-06-22	2	Visa	Delivered	15
8	8	34	10	2023-12-28	4	Cash	Delivered	0
9	9	7	7	2023-02-16	1	Instapay	Delivered	5
10	10	68	14	2023-05-21	3	Instapay	Delivered	15
11	11	58	15	2023-04-28	2	Visa	Delivered	0
12	12	75	5	2023-10-04	4	Instapay	Delivered	20
13	13	29	7	2023-08-02	1	Visa	Delivered	0
14	14	36	9	2023-08-08	1	Visa	Delivered	0
15	15	89	3	2023-01-26	3	Cash	Delivered	15
16	16	21	8	2023-01-09	4	Cash	Delivered	0
17	17	36	2	2023-09-28	3	Visa	Cancelled	20

```sql
select *
from orders
where year(order_date) = 2023
```

Q2 - Retrieve all orders with status = 'Delivered'.

Results | Messages

	order_id	customer_id	product_id	order_date	quantity	payment_method	order_status	discount
1	1	81	10	2023-10-25	3	Instapay	Delivered	0
2	2	90	7	2023-02-05	1	Cash	Delivered	5
3	3	8	1	2023-03-25	3	Cash	Delivered	0
4	4	93	7	2023-09-20	2	Cash	Delivered	10
5	6	74	5	2023-06-21	1	Cash	Delivered	0
6	7	90	6	2023-06-22	2	Visa	Delivered	15
7	8	34	10	2023-12-28	4	Cash	Delivered	0
8	9	7	7	2023-02-16	1	Instapay	Delivered	5
9	10	68	14	2023-05-21	3	Instapay	Delivered	15
10	11	58	15	2023-04-28	2	Visa	Delivered	0
11	12	75	5	2023-10-04	4	Instapay	Delivered	20
12	13	29	7	2023-08-02	1	Visa	Delivered	0
13	14	36	9	2023-08-08	1	Visa	Delivered	0
14	15	89	3	2023-01-26	3	Cash	Delivered	15
15	16	21	8	2023-01-09	4	Cash	Delivered	0
16	18	10	11	2023-01-10	3	Visa	Delivered	20
17	19	73	7	2023-05-01	4	Cash	Delivered	0

```sql
select *
from orders
where order_status = 'Delivered'
```

Q3 - Retrieve all customers who live in Cairo.

	customer_id	customer_name	city	age	join_date
1	15	Mona Adel	Cairo	52	2023-06-16
2	16	Dina Mostafa	Cairo	58	2021-09-29
3	19	Ibrahim Adel	Cairo	46	2022-03-23
4	22	Ibrahim Samir	Cairo	50	2023-05-11
5	23	Mohamed Ibrahim	Cairo	38	2021-09-04
6	33	Mona Ali	Cairo	57	2021-02-16
7	36	Yara Adel	Cairo	39	2022-03-29
8	38	Aya Ali	Cairo	19	2021-08-06
9	45	Heba Fathy	Cairo	50	2021-09-11
10	46	Mahmoud Mahmoud	Cairo	26	2023-01-04
11	52	Ahmed Adel	Cairo	42	2023-04-17
12	55	Mahmoud Hassan	Cairo	24	2021-06-29
13	62	Ali Hassan	Cairo	29	2021-11-21
14	64	Omar Adel	Cairo	32	2021-01-16
15	73	Ahmed Mostafa	Cairo	39	2023-05-02
16	75	Mahmoud Adel	Cairo	23	2022-11-21
17	77	Ibrahim Saeed	Cairo	54	2022-01-15

```
select *
from customers
where city = 'Cairo'
```

Q4 - Retrieve all products with price greater than 5000.

	product_id	product_name	category	price
1	1	Laptop	Electronics	20000
2	2	iPhone	Electronics	30000
3	3	Samsung Phone	Electronics	15000
4	7	Monitor	Electronics	6000
5	8	Tablet	Electronics	10000
6	10	Camera	Electronics	12000

```
select *
from products
where price > 5000
```

Q5 - Retrieve all orders where the discount is greater than 10%.

	order_id	customer_id	product_id	order_date	quantity	payment_method	order_status	discount
1	5	26	10	2023-11-06	4	Cash	Pending	15
2	7	90	6	2023-06-22	2	Visa	Delivered	15
3	10	68	14	2023-05-21	3	Instapay	Delivered	15
4	12	75	5	2023-10-04	4	Instapay	Delivered	20
5	15	89	3	2023-01-26	3	Cash	Delivered	15
6	17	36	2	2023-09-28	3	Visa	Cancelled	20
7	18	10	11	2023-01-10	3	Visa	Delivered	20
8	27	96	2	2023-12-13	4	Cash	Delivered	20
9	34	55	8	2023-03-01	1	Cash	Delivered	20
10	38	99	13	2023-05-26	2	Cash	Cancelled	15
11	45	13	7	2023-08-14	2	Instapay	Delivered	15
12	59	52	13	2023-11-16	2	Visa	Delivered	20
13	62	69	14	2023-03-26	1	Visa	Delivered	15
14	63	99	2	2023-05-26	4	Cash	Delivered	20
15	79	95	13	2023-04-19	3	Cash	Delivered	20
16	82	87	13	2023-08-17	1	Cash	Pending	20
17	85	18	12	2023-03-14	2	Cash	Delivered	15

```
select *
from orders
where discount > 10
```

Q6 - Count the total number of orders.

	Total_Orders
1	1000

```
select COUNT(*) Total_Orders
from orders
```

Q7 - Calculate the total quantity of products sold.

	total_quantity_sold
1	2478

```
select SUM(quantity) total_quantity_sold
from orders
```

Q8 - Calculate the average product price.

	Avg_Price
1	7326

```
select AVG(price) Avg_Price
from products
```

Q9 - Count the number of orders for each order status (Delivered, Cancelled, Pending).

	order_status	Orders_Count
1	Pending	86
2	Delivered	815
3	Cancelled	99

```
select order_status, count(*) Orders_Count
from orders
group by order_status
```

Q10 - Count the number of customers in each city.

	city	Customer_Count
1	Alexandria	21
2	Cairo	19
3	Giza	10
4	Ismailia	9
5	Mansoura	18
6	Tanta	7
7	Zagazig	16

```
select city, count(*) Customer_Count
from customers
group by city
```

Q11 - Retrieve each order along with the customer name who made it.

	order_id	customer_name
1	1	Ibrahim Ibrahim
2	2	Ahmed Mahmoud
3	3	Mohamed Fathy
4	4	Youssef Adel
5	5	Fatma Mahmoud
6	6	Dina Mostafa
7	7	Ahmed Mahmoud
8	8	Tarek Ibrahim
9	9	Mahmoud Khaled
10	10	Mohamed Samir
11	11	Ahmed Saeed
12	12	Mahmoud Adel
13	13	Hassan Ibrahim
14	14	Yara Adel
15	15	Omar Hassan
16	16	Yara Ali
17	17	Yara Adel

```
select o.order_id, c.customer_name
from orders o
join customers c
on o.customer_id = c.customer_id
```

Q12 - Retrieve each product along with the total quantity sold.

	product_name	Total_Quantity
1	Camera	169
2	Charger	156
3	Desk Lamp	169
4	Headphones	131
5	iPhone	219
6	Keyboard	152
7	Laptop	172
8	Monitor	152
9	Mouse	143
10	Office Chair	174
11	Printer	184
12	Samsung Phone	152
13	Smart Watch	168
14	Speaker	190
15	Tablet	147

```
select p.product_name, sum(o.quantity) as
Total_Quantity
from orders o
join products p
on o.product_id = p.product_id
group by p.product_name
```

Q13 - Calculate total sales for each product (price × quantity × (1 - discount/100))

	product_name	Total_Sales
1	Desk Lamp	152100
2	Laptop	3440000
3	Office Chair	435000
4	Monitor	912000
5	Mouse	57200
6	Speaker	475000
7	Charger	46800
8	Smart Watch	840000
9	Keyboard	152000
10	Camera	2028000
11	Tablet	1470000
12	iPhone	6570000
13	Samsung Phone	2280000
14	Headphones	104800
15	Printer	644000

```
select p.product_name, SUM(p.price *
o.quantity * (1-o.discount/100)) Total_Sales
from orders o
join products p
on o.product_id = p.product_id
group by p.product_name
```

Q14 - Find the top 5 customers based on total spending.

Results | Messages

	customer_name	Total_Sales
1	Mona Ali	660400
2	Sara Fathy	635300
3	Yara Adel	577300
4	Dina Mostafa	570700
5	Mona Fathy	566900

```
select top(5) c.customer_name, SUM(p.price *
o.quantity * (1-o.discount/100)) Total_Sales
from orders o
join products p
on o.product_id = p.product_id
join customers c
on o.customer_id = c.customer_id
group by c.customer_name
order by Total_Sales desc
```

Q15 - Retrieve customers who made more than 3 orders.

	customer_name	orders_count
1	Sara Fathy	26
2	Mostafa Fathy	11
3	Heba Hassan	21
4	Mohamed Ibrahim	12
5	Mohamed Khaled	19
6	Omar Ibrahim	10
7	Fatma Hassan	5
8	Tarek Fathy	6
9	Mahmoud Khaled	8
10	Mona Samir	16
11	Heba Saeed	12
12	Youssef Ibrahim	10
13	Omar Adel	10
14	Mahmoud Hassan	11
15	Mohamed Ali	8
16	Heba Fathy	22
17	Ahmed Samir	8

```sql
select c.customer_name, COUNT(*) orders_count
from orders o
join customers c
on o.customer_id = c.customer_id
group by c.customer_name
having(COUNT(*) > 3)
```

Q16 - Find the total sales for each city.

	city	Total_Sales
1	Zagazig	3443300
2	Giza	1440000
3	Cairo	3660200
4	Mansoura	3230700
5	Alexandria	4654800
6	Tanta	1211700
7	Ismailia	1966200

```
select c.city, SUM(p.price * o.quantity *
(1-o.discount/100)) Total_Sales
from orders o
join products p
on o.product_id = p.product_id
join customers c
on o.customer_id = c.customer_id
group by c.city
```

Q17 - Find the most sold product (by quantity).

	product_name	Total_Quantity
1	iPhone	219

```
select top(1) p.product_name, sum(o.quantity)
Total_Quantity
from orders o
join products p
on o.product_id = p.product_id
group by p.product_name
order by Total_Quantity desc
```

Q18 - Calculate the total revenue generated for each payment method.

	payment_method	Total_Sales
1	Instapay	2134500
2	Visa	8216000
3	Cash	9256400

```
select o.payment_method, SUM(p.price *
o.quantity * (1-o.discount/100)) Total_Sales
from orders o
join products p
on o.product_id = p.product_id
group by o.payment_method
```

Q19 - Find the monthly total sales trend.

	Month_Name	Total_Sales
1	April	1550400
2	August	1332100
3	December	1430100
4	February	1532800
5	January	1602900
6	July	1302100
7	June	2247500
8	March	1995800
9	May	1817500
10	November	1427100
11	October	1286700
12	September	2081900

```
select datename(MONTH, o.order_date) as
Month_Name, SUM(p.price * o.quantity * (1-
o.discount/100)) Total_Sales
from orders o
join products p
on o.product_id = p.product_id
group by datename(MONTH, o.order_date)
```

Q20 - Identify high-value customers who: made more than 3 orders and spent more than 20000

	customer_name	orders_count	Total_Sales
1	Ahmed Adel	8	49100
2	Ahmed Fathy	5	257600
3	Ahmed Mahmoud	23	462400
4	Ahmed Mostafa	10	95400
5	Ahmed Saeed	22	315200
6	Ahmed Samir	8	244100
7	Ali Hassan	24	403400
8	Ali Ibrahim	22	454600
9	Ali Mostafa	8	248900
10	Ali Samir	9	171300
11	Aya Ali	16	261600
12	Aya Mahmoud	10	166000
13	Dina Adel	9	205800
14	Dina Ibrahim	12	227300
15	Dina Mostafa	22	570700
16	Fatma Adel	9	245200
17	Fatma Fathy	8	145600

```
select c.customer_name, COUNT(*) orders_count, SUM(p.price *
o.quantity * (1-o.discount/100)) Total_Sales
from orders o
join products p
on o.product_id = p.product_id
join customers c
on o.customer_id = c.customer_id
group by c.customer_name
having(COUNT(*) > 3) and SUM(p.price * o.quantity * (1-
o.discount/100)) > 20000
```

Summary

- **81% of orders are successfully delivered**, indicating strong operational performance.
- On the other hand, **the 10 % cancelled orders** represents a noticeable loss in potential revenue.
- **About 9% of orders are pending** and that may indicate delays in processing or fulfillment.
- **Most orders have no discount** ,while some reach up to 20%.
- **Cash** is the dominant payment method **(47.3%)**, followed by **Visa** (42.3%).

Recommendations

- Improve order confirmation and delivery processes to recover **10% lost orders.**
- Investigate delays in processing to improve customer experience.
- Use targeted discounts instead of random usage.
- Offer incentives like (discounts or cashback) to reduce reliance on cash.

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in Mohamed Hassan